

Tomaz Gonçalves Silva

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<https://portfolio-tomaz.netlify.app/>

Education

Technical University of Denmark

MSC IN COMPUTER SCIENCE, SPECIALISING IN **CYBER SECURITY**

GPA: 12/12

Copenhagen

2024-2028

Instituto Superior Técnico

BSC IN COMPUTER SCIENCE

GPA: 3.3 / 4

Lisbon

Sep 2021 – Jun 2024

Work Experience

Cloud Engineer (Full-Time Internship)

Copenhagen (UN City)

UNITED NATIONS (UNDP)

August 2025 – Current

- Managed and deployed cloud resources (**VMs**, **ContainerApps**, etc.) in **Azure** for 100+ subscriptions serving global UN country offices.
- Implemented and maintained **Kubernetes** clusters to run scalable microservice-based applications in the cloud.
- Configured and managed **RBAC** permissions to resources and **Entra ID** (Active Directory) users to strengthen access control and security.
- Monitored thousands of enrolled devices with **Microsoft Intune** to ensure policy compliance and secure configurations.
- Collaborated with international teams to design and deliver cloud solutions fitting their requirements.

Backend Engineer (Student position)

Copenhagen

BIO SPRINGBOK APS

September 2024 – May 2025

- Architected and implemented an autonomous pipeline in AWS cloud, using **S3** for encrypted large data storage, **Lambda** for event driven processing and **EC2** for deployment.
- Created clusters, following **IAC** principles, with **AWS ParallelCluster** to parallelize heavy machine learning work.
- Automated cloud infrastructure with **Bash** and **Python** to streamline DevOps and reduce deployment time.
- Managed a **PostgreSQL** database hosted on a cloud **Linux** machine with security protocols in place for sensitive medical data.
- Developed backend using **Spring Boot** and **Java 21**, integrating AWS SDK connectors.

Machine Learning Intern

Lisbon

INOVA (HTTPS://WWW.CICLOPE.COM.PT/)

March 2024 – October 2024

- Trained and evaluated multiple classifiers to automatically detect wildfires in nature using **Python** and **PyTorch**.
- Preprocessed thousands of images with **OpenCV** and **Pandas**, including data augmentation and cleaning for robust model performance.
- Developed an **end-to-end pipeline** to compare configurations and obtain performance metrics, achieving **91% accuracy** with only 5% false positives.

Projects

DHCP Spoofing and MITM Attack (Personal)

C, DHCP, DNS, VMs

HTTPS://GITHUB.COM/TOMHOQ/DHCP-SPOOFING-AND-MITM-ATTACK

2025

- Implemented a DHCP starvation attack and a simulated rogue DHCP server in **C** to demonstrate MITM and DNS manipulation techniques.
- Configured a test environment with virtual machines and DNS settings to safely simulate network vulnerabilities.
- Documented challenges and mitigation strategies to improve my understanding of DHCP/DNS security and network protocols.
- Applied raw sockets and randomized MAC addresses to counter defense techniques, emphasizing low-level network programming and security concepts.

Skills

Programming Languages: Java, Python, C, C++, SQL, JavaScript, TypeScript, Bash, Powershell

Libraries & Frameworks: Spring Boot, React, Vue, Flask, Pytorch, Scikit-Learn

Cloud & DevOps: AWS (EC2, Lambda, S3), Git, Docker, Kubernetes (basic)

Languages: Portuguese (Native), English (C2), Danish (A2)

Hobbies & Awards

2022– **Self-taught Guitar and Piano Player,**

2019-2021 **Academic Merit,** GPA 3.9 / 4

2016– **Casual Tennis Player,**

2005-2024 **Roller Hockey Goalkeeper,**

Escola Dr. António Carvalho Figueiredo

Atlético Clube do Tojal